Ej.: $\begin{cases} 2x - y = 2 \\ x + z = 0 \\ -y - 2z = 2 \end{cases} \longrightarrow \underbrace{\begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix}}_{A} \cdot \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}}_{b}$

$A \cdot x = b$

SEA $v = \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

CALCULAR $A.v$, $A.w$, $A.(v+w)$.

S)

$$A.v = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \longrightarrow v \text{ ES SOLUCIÓN DE } Ax = b$$

$$A.w = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \longrightarrow w \text{ ES SOLUCIÓN DEL SISTEMA HOMOGÉNEO ASOCIADO } (Ax = 0)$$

$$A.(v+w) = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix} \cdot \left(\begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$$

OBS.: $A.(v+w) = \underbrace{Av}_{b} + \underbrace{Aw}_{0} = b + 0 = b$

OBS.: S_1 $S_0 = \{x : Ax = 0\}$, $S_b = \{x : Ax = b\}$ y $b \in S_b$,

ENTONCES

$$S_b = v + S_0 \overset{d}{=} \{v + w : w \in S_0\}$$

- SI $v \in S_b$, $w \in S_0 \Rightarrow v + w \in S_b$

$(A(v+w) = Av + Aw = b + 0 = b)$

- SI $v, u \in S_b$, ENTONCES $u = v + u - v$, PERO $u - v \in S_0$. EN EFECTO, $A(u-v) = A.u - Av = b - b = 0$.

$\therefore u \in v + S_0$.

MATRIZ INVERSA:

DEF.: SEA $A \in \mathbb{R}^{n \times n}$. A ES INVERSIBLE SI EXISTE $B \in \mathbb{R}^{n \times n}$ TAL QUE $A \cdot B = B \cdot A = I_n$.

NOTACIÓN: $B = A^{-1}$ → MATRIZ INVERSA DE A.

EJ: HALLAR LA MATRIZ INVERSA DE $A = \begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix}$.

S/ DADA A, SE BUSCA $X = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}$ TAL QUE

$$A \cdot X = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} = \begin{bmatrix} 8x_1 + 3y_1 & 8x_2 + 3y_2 \\ 5x_1 + 2y_1 & 5x_2 + 2y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$8x_1 + 3y_1$ y $5x_1 + 2y_1$ = $A \cdot \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$

$8x_2 + 3y_2$ y $5x_2 + 2y_2$ = $A \cdot \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$

ESTO EQUIVALE A

$$\begin{cases} 8x_1 + 3y_1 = 1 \\ 5x_1 + 2y_1 = 0 \end{cases} \quad \text{y} \quad \begin{cases} 8x_2 + 3y_2 = 0 \\ 5x_2 + 2y_2 = 1 \end{cases}$$

$$\left[\begin{array}{cc|c} 8 & 3 & 1 \\ 5 & 2 & 0 \end{array}\right] \qquad \left[\begin{array}{cc|c} 8 & 3 & 0 \\ 5 & 2 & 1 \end{array}\right]$$

∴ PODEMOS PLANTEAR Y RESOLVER DE FORMA SIMULTÁNEA:

$$\left[\begin{array}{cc|cc} 8 & 3 & 1 & 0 \\ 5 & 2 & 0 & 1 \end{array}\right] \xrightarrow{2F_1 - 3F_2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 5 & 2 & 0 & 1 \end{array}\right] \xrightarrow{F_2 - 5F_1} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 2 & -10 & 16 \end{array}\right] \xrightarrow{\frac{1}{2}F_2}$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 1 & -5 & 8 \end{array}\right] \Longrightarrow \boxed{A^{-1} = \begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix}}$$

I $\qquad$ A^{-1}

OBS.: $A \in \mathbb{R}^n$ ES INVERSIBLE $\iff$ $\text{rg}(A) = n$.

EJEMPLOS:

1) $A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 2 & -1 & 1 & 0 \end{array}\right] \xrightarrow{F_2 - 2F_1} \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & -7 & 1 & -2 \end{array}\right] \xrightarrow{-\frac{1}{7}F_2} \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & 1 & -\frac{1}{7} & \frac{2}{7} \end{array}\right]$$

$$\xrightarrow{F_1 - 3F_2} \left[\begin{array}{cc|cc} 1 & 0 & \frac{3}{7} & \frac{1}{7} \\ 0 & 1 & -\frac{1}{7} & \frac{2}{7} \end{array}\right] \Longrightarrow A^{-1} = \begin{bmatrix} \frac{3}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{2}{7} \end{bmatrix}.$$

2) $B = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 4 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array}\right] \xrightarrow{F_2 - \frac{1}{2}F_1} \left[\begin{array}{cc|cc} 4 & 2 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{array}\right]$$

$\therefore$ B NO ES INVERSIBLE

$\left[\begin{array}{cc|c} 4 & 2 & 0 \\ 0 & 0 & 1 \end{array}\right]$ → SISTEMA INCOMPATIBLE.

$\left[\begin{array}{cc|c} 4 & 2 & 1 \\ 0 & 0 & -\frac{1}{2} \end{array}\right]$ → SISTEMA INCOMPATIBLE

3) $C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[F_3 - 3F_1]{F_2 - 2F_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & -1 & -2 & 1 & 0 \\ 0 & 1 & -1 & -3 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -3 & 0 & 1 \\ 0 & 2 & -1 & -2 & 1 & 0 \end{array}\right] \xrightarrow{F_3 - 2F_2}$$

$$\longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -3 & 0 & 1 \\ 0 & 0 & 1 & 4 & 1 & -2 \end{array}\right] \xrightarrow[F_2 + F_3]{F_1 - F_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & -1 & 2 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 4 & 1 & -2 \end{array}\right] \Rightarrow C^{-1} = \begin{bmatrix} -3 & -1 & 2 \\ 1 & 1 & -1 \\ 4 & 1 & 2 \end{bmatrix}$$

EJEMPLO: SEA $A = \begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix}$ Y SEA EL SISTEMA

$A \cdot X = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. VIMOS QUE $A^{-1} = \begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix}$.

$$AX = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$A^{-1} \cdot (AX) = A^{-1} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\underbrace{(A^{-1}.A)}_{=} . X = A^{-1}.\begin{bmatrix}1\\3\end{bmatrix}$$

$$I_2 . X = A^{-1}.\begin{bmatrix}1\\3\end{bmatrix}$$

$$X = A^{-1}.\begin{bmatrix}1\\3\end{bmatrix} = \begin{bmatrix}2 & -3\\ -5 & 8\end{bmatrix}\begin{bmatrix}1\\3\end{bmatrix}$$

$$\boxed{X = \begin{bmatrix}-7\\19\end{bmatrix}}$$

NOTA: EL SISTEMA $Ax = b$ ($A \in \mathbb{R}^{n\times n}$, $b \in \mathbb{R}^{n\times 1}$) TIENE SOLUCIÓN ÚNICA SI Y SÓLO SI A ES INVERSIBLE. EN TAL CASO, LA SOLUCIÓN ES $\boxed{X = A^{-1}.b}$.

OBS.: SI $A, B \in \mathbb{R}^{n\times n}$ SON INVERSIBLES, ENTONCES $(AB)^{-1} = B^{-1}.A^{-1}$.

$$AB.(B^{-1}.A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} =$$

$$= A.A^{-1} = I.$$

$$B^{-1}.A^{-1}.AB = I$$